WORKED SOLUTION

STEP I 1989, Question 9

Pure • Transformations of Graphs

STEP I 1989, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q9

The key idea

Once the basic cubic is sketched, every other graph is the same curve seen through a change of coordinates. A substitution $X = ax + b$, $Y = cy + d$ is just a stretch (or reflection) in each direction followed by a translation, and it carries turning points to turning points. So all the work is in finding the two turning points of the original cubic and tracking where they go.

Solution

The basic graph

Write $y = \frac{1}{8}(x^3 - 12x) = \frac{1}{8}x(x^2 - 12)$. Then

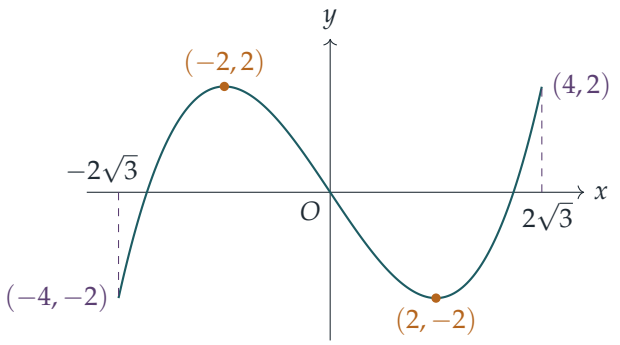
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3x^2 - 12}{8} = \frac{3(x-2)(x+2)}{8}, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{6x}{8} = \frac{3x}{4}.$$

The gradient is zero at $x = \pm 2$.

- At $x = -2$: $y = \frac{1}{8}(-8 + 24) = 2$, and $y'' = -\frac{3}{2} < 0$, so $(-2, 2)$ is a local maximum.
- At $x = 2$: $y = \frac{1}{8}(8 - 24) = -2$, and $y'' = \frac{3}{2} > 0$, so $(2, -2)$ is a local minimum.

Other useful features:

- The curve meets the axes where $x(x^2 - 12) = 0$, i.e. at $x = 0$ and $x = \pm 2\sqrt{3} \approx \pm 3.46$.
- The function is odd, so the graph has rotational symmetry of order 2 about the origin, which is also the point of inflection; the gradient there is $-\frac{3}{2}$.
- At the ends, $y(4) = \frac{1}{8}(64 - 48) = 2$ and $y(-4) = -2$. So the curve climbs back to the height of the maximum exactly at $x = 4$.



Why turning points go to turning points

Ask yourself When we change coordinates, how do we know the new turning points are just the images of the old ones?

If $X = ax + b$ and $Y = cy + d$ with $a, c \neq 0$, then by the chain rule

$$\frac{\mathrm{d}Y}{\mathrm{d}X} = \frac{\mathrm{d}Y/\mathrm{d}x}{\mathrm{d}X/\mathrm{d}x} = \frac{c}{a}\frac{\mathrm{d}y}{\mathrm{d}x},$$

which is zero exactly where $\mathrm{d}y/\mathrm{d}x$ is zero. So the new turning points are the images of $(-2, 2)$ and $(2, -2)$. (If $c/a < 0$ the picture is reflected and the maximum and minimum swap roles.) Geometrically:

- $X = \frac{1}{2}x$ squashes the graph horizontally by a factor of 2;
- $Y = \frac{1}{2}y$ squashes it vertically by a factor of 2;
- adding 1 to X or Y then shifts it 1 unit right or up.

The four transformed graphs

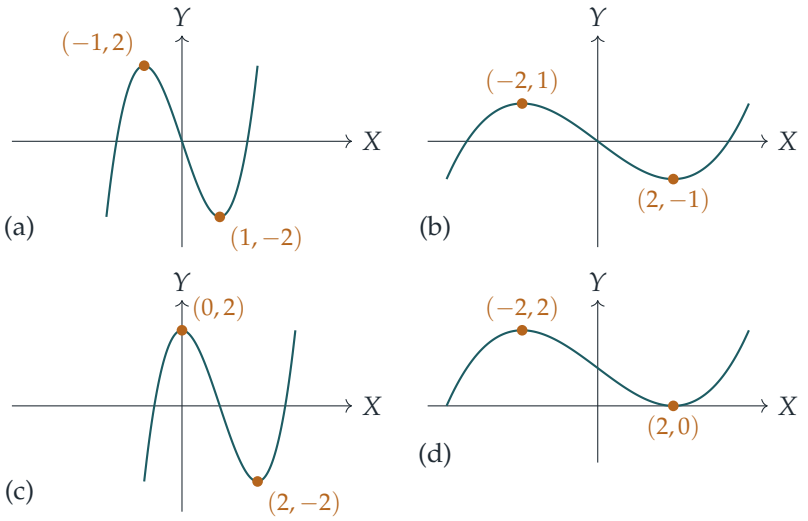
Applying each map to $(-2, 2)$, $(2, -2)$ and to the range $-4 \leq x \leq 4$:

(a) $X = \frac{1}{2}x$, $Y = y$: maximum $(-1, 2)$, minimum $(1, -2)$, for $-2 \leq X \leq 2$. The curve is $Y = X^3 - 3X$.

(b) $X = x$, $Y = \frac{1}{2}y$: maximum $(-2, 1)$, minimum $(2, -1)$, for $-4 \leq X \leq 4$. The curve is $16Y = X^3 - 12X$.

(c) $X = \frac{1}{2}x + 1$, $Y = y$: maximum $(0, 2)$, minimum $(2, -2)$, for $-1 \leq X \leq 3$. This is graph (a) moved 1 unit right, with inflection at $(1, 0)$.

(d) $X = x$, $Y = \frac{1}{2}y + 1$: maximum $(-2, 2)$, minimum $(2, 0)$, for $-4 \leq X \leq 4$. This is graph (b) moved 1 unit up, with inflection at $(0, 1)$; it touches the X-axis at its minimum.



The final part

We need $a, c \neq 0$ (otherwise the graph collapses to a line), and then the turning points of the new graph are

$$(-2a + b,\ 2c + d) \quad \text{and} \quad (2a + b,\ -2c + d).$$

These must be $(0, 0)$ and $(1, 1)$ in some order, which gives two linear systems.

Case 1: $(-2, 2) \mapsto (0, 0)$ and $(2, -2) \mapsto (1, 1)$. Then $-2a + b = 0$ and $2a + b = 1$, so $b = \frac{1}{2}$, $a = \frac{1}{4}$; and $2c + d = 0$, $-2c + d = 1$, so $d = \frac{1}{2}$, $c = -\frac{1}{4}$.

Case 2: $(-2, 2) \mapsto (1, 1)$ and $(2, -2) \mapsto (0, 0)$. In the same way $a = -\frac{1}{4}$, $b = \frac{1}{2}$, $c = \frac{1}{4}$, $d = \frac{1}{2}$.

So one answer is

$$a = \tfrac{1}{4}, \quad b = \tfrac{1}{2}, \quad c = -\tfrac{1}{4}, \quad d = \tfrac{1}{2},$$

and the other is $a = -\frac{1}{4}$, $b = \frac{1}{2}$, $c = \frac{1}{4}$, $d = \frac{1}{2}$.

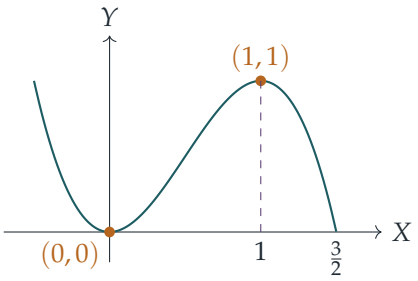
It is worth finding the new curve. In Case 1, $x = 4X - 2$ and $y = 2 - 4Y$, so

$$\begin{aligned} 8(2 - 4Y) &= (4X - 2)^3 - 12(4X - 2) \\ &= 64X^3 - 96X^2 + 48X - 8 - 48X + 24 \\ &= 64X^3 - 96X^2 + 16, \end{aligned}$$

which simplifies to

$$Y = 3X^2 - 2X^3 = X^2(3 - 2X).$$

This has $Y' = 6X(1 - X)$, so its turning points are indeed a minimum at $(0, 0)$ and a maximum at $(1, 1)$. Case 2 gives exactly the same curve: replacing (x, y) by $(-x, -y)$ doesn't change the original graph, because the cubic is odd, and that's the only difference between the two cases.



For $-4 \leq x \leq 4$ this runs from $(-\frac{1}{2}, 1)$ to $(\frac{3}{2}, 0)$.

Checks

- Substituting $X = 0$ and $X = 1$ into $Y = 3X^2 - 2X^3$ gives 0 and 1, and $Y' = 6X(1 - X)$ vanishes at both.
- Each transformed graph keeps the original feature that the curve returns to the height of a turning point at an end of the range: in (d), for instance, $Y = 2$ at $X = 4$ and $Y = 0$ at $X = -4$.
- Numerically, at $x = 1$ we have $y = -\frac{11}{8}$; in Case 1 this gives $X = 0.75$, $Y = 0.84375$, and $3(0.75)^2 - 2(0.75)^3 = 0.84375$.

Common slips

- Getting the direction of a stretch wrong: $X = \frac{1}{2}x$ makes the graph *narrower*, not wider, because the point at $x = 2$ moves to $X = 1$.
- Forgetting to transform the domain as well: graph (c) runs over $-1 \leq X \leq 3$, not $-4 \leq X \leq 4$.
- Giving only one solution to the final part without noticing the other ordering of the turning points (either is enough, but $c < 0$ in the first case is easy to miss).

Beyond STEP

The maps $X = ax + b$, $Y = cy + d$ are affine transformations, and the final part shows that any cubic with two turning points can be moved by one of them onto the standard curve $Y = 3X^2 - 2X^3$; this is the "smoothstep" function used in computer graphics. Classifying curves up to a group of transformations like this is a theme that runs through geometry and linear algebra. For the matrix side of the story, see Strang, *Introduction to Linear Algebra*.

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